

TW14: Supervised Machine Learning

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Build leakage-safe classification and regression workflows, compare required algorithms, and select metrics according to business error costs.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Prepare data and establish a baseline

Scenario

A retention team wants to predict churn, but the target is imbalanced and customer IDs must not become predictors.

INSTRUCTIONS

1. Import customer_churn.csv and audit target balance.
2. Separate identifier, feature matrix, and target.
3. Create a 75/25 stratified split with random_state 42.
4. Build numeric and categorical preprocessing.
5. Evaluate a most-frequent DummyClassifier using accuracy, precision, recall, and F1.

REQUIRED SUBMISSION

- A validated split and preprocessing object.
- A baseline metric table.
- An explanation of why baseline accuracy can mislead.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Train logistic regression and tune a decision threshold

Scenario

The business values recall but has limited outreach capacity.

INSTRUCTIONS

1. Fit logistic regression in a complete preprocessing pipeline.
2. Report confusion matrix, precision, recall, F1, accuracy, and ROC AUC on test data.
3. Compare thresholds 0.30, 0.50, and 0.70.
4. Recommend a threshold using false-negative cost and predicted-positive volume.

REQUIRED SUBMISSION

- A fitted leakage-safe model.
- A threshold comparison table and confusion matrix.
- A decision memo tied to operational capacity.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Compare decision tree and k-NN

Scenario

The team wants to compare an interpretable rule model with a neighbour-based model.

INSTRUCTIONS

1. Fit a constrained decision tree with a fixed random seed.
2. Fit scaled k-NN models for at least three values of k.
3. Compare all candidates on the same test set using recall, F1, and ROC AUC where available.
4. Discuss overfitting, scaling, and interpretability.

REQUIRED SUBMISSION

- A model-comparison table.
- Evidence supporting tree complexity and chosen k.
- A reasoned candidate selection rather than a largest-score-only answer.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Cross-validate LDA, GaussianNB, and SVC

Scenario

A single split may be unstable, so three additional classifiers require repeated fold evidence.

INSTRUCTIONS

1. Create dense one-hot preprocessing compatible with all three algorithms.
2. Use five-fold StratifiedKFold with shuffle and random_state 42.
3. Report mean and standard deviation for precision, recall, F1, and accuracy.
4. Compare training and validation scores for the selected candidate.
5. Preserve a final holdout for confirmation.

REQUIRED SUBMISSION

- A complete cross-validation results table.
- A variability and overfitting discussion.
- A candidate recommendation with final-test protocol.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Complete a regression walkthrough

Scenario

A chain wants to predict continuous weekly revenue from store characteristics.

INSTRUCTIONS

1. Generate or construct at least 200 reproducible rows with four predictors and a noisy continuous target.
2. Create a train/test split.
3. Fit linear regression and constrained decision-tree regression.
4. Report MAE, RMSE, and R-squared on identical test rows.
5. Run five-fold cross-validation and inspect residual patterns.

REQUIRED SUBMISSION

- A reproducible regression dataset.
- Holdout and cross-validation comparison tables.
- A residual chart and model recommendation with limitations.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____